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Preserving Archival Heritage: A Machine Learning Model Language for Spanish Transcripts

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ABSTRACT

Historical archives from the Philippine-American War (1899–1902) hold vital records of the nation's past, yet much of this heritage remains inaccessible because it is written by hand in period Spanish, featuring archaic spellings, abbreviations, and degraded paper quality. Standard digitization captures images but fails to provide semantic meaning, requiring rare paleographic expertise. To address this problem, this study aims to develop and evaluate an automated computational pipeline to convert scanned historical transcripts into readable, searchable modern English text. The proposed approach implements a three-stage framework: (1) advanced image preprocessing using OpenCV for grayscale conversion, noise removal, contrast enhancement, deskewing, and thresholding; (2) text recognition using Tesseract OCR and TrOCR; and (3) neural machine translation via a pretrained Marian model coupled with rule-based spelling modernization. The intended dataset comprises an estimated 44 high-resolution scanned manuscript images acquired through collaboration with the University of Nueva Caceres Museum. The expected contribution of this work is a documented, end-to-end workflow that empowers historians, archivists, and the public to efficiently uncover, transcribe, and translate forgotten archival records, supporting the long-term preservation and accessibility of historical heritage.

1. KEYWORDS

Machine translation, historical Spanish, digital image processing, optical character recognition, OpenCV

2. INTRODUCTION

2.1 Background of the Study

Historical archives preserve the written record of how societies lived, governed and fought, and for the Philippines a large part of that record is written in Spanish. Spanish was the language of colonial administration until 1898, and it remained the working language of the Philippine Revolution and the First Philippine Republic, including its constitution, correspondence and military orders. During the Philippine-American War (1899–1902), many of these documents were written by hand, often in the field and on poor paper, in a style that differs from modern Spanish in spelling, abbreviations, formal expressions and vocabulary. Collections such as the Philippine Insurgent Records hold many of these papers, and archives increasingly digitize them as scanned images to protect the originals.

Digitization, however, saves only the picture of a document and not its meaning. A scanned page must still be read, transcribed and translated by someone trained in both paleography and period Spanish, and such experts are few. Standard commercial translation tools are trained mostly on modern, printed text, so they handle archaic spelling, wartime terminology and degraded handwriting poorly. Machine learning, combined with image processing libraries such as OpenCV, offers a way to turn these scans into readable, searchable and translated text, so that the records of this period become accessible to historians, students and the general public.

2.2 Problem Context

Even when the text is recovered, it is written in period Spanish with old spellings and abbreviations, so general-purpose translators often produce errors or unreadable output. As a result, the people, deeds and sacrifices recorded in these documents stay locked inside images

that few can read, and the history they hold risks going unnoticed.

2.3 Research Motivation

The researchers are motivated by the need to keep this history from going unnoticed. By building a machine learning model that takes a scanned transcript image, cleans it with OpenCV, recognizes its text and translates it into modern English, the study aims to give historians a faster way to recover records that would otherwise remain unread, and to help the public relearn and recognize the events and people of that period.

2.4 Research Questions

The study follows a three-stage pipeline: (1) image preprocessing with OpenCV, (2) text recognition (OCR/HTR) on the cleaned image, and (3) translation of the recognized Spanish text. The questions below follow those stages.

- **RQ1:** How do OpenCV preprocessing techniques (grayscale conversion, noise removal, CLAHE contrast enhancement and deskewing) affect the legibility of degraded Philippine-American War manuscript images?
- **RQ2:** Which thresholding technique (global, Otsu or adaptive) gives the clearest separation between handwritten text and the background in these images?
- **RQ3:** How accurately can an OCR engine such as Tesseract with Spanish language data recognize text from the preprocessed images, measured by character error rate and word error rate?
- **RQ4:** How accurately can a machine translation model fine-tuned on period Spanish translate the recognized text into modern English, measured by BLEU score and human review by historians?

2.5 Objectives

2.5.1 General Objective

To develop a machine learning model that reads scanned images of Spanish manuscripts from the Philippine-American War period and translates them into modern English, so that historians can access and preserve these records.

2.5.2 Specific Objectives

1. To build an OpenCV preprocessing pipeline (grayscale conversion, denoising, CLAHE and deskewing) that improves the legibility of degraded manuscript images.

2. To compare global, Otsu and adaptive thresholding and select the one that best separates handwritten text from the background.

3. To integrate an OCR engine that extracts Spanish text from the preprocessed images and to measure its accuracy using character and word error rates.

4. To fine-tune a machine translation model on period Spanish text and its English translations so that it handles archaic spelling and vocabulary.

5. To evaluate the complete image-to-translation workflow using BLEU score and the feedback of historians or Spanish-language experts.

2.6 Significance of the Study

Historians and researchers gain a faster way to read and translate manuscripts of the Philippine-American War, which lets them study more documents and recover details of events and people that were previously unread.

Archivists and librarians gain a method to turn scanned collections into searchable, translated text, which supports the long-term preservation of fragile records.

Students and the general public gain access to primary sources in a language they can read, which helps them learn and recognize the sacrifices and deeds recorded in the nation's history.

Future researchers gain a documented pipeline and evaluation results that can be extended to other periods, handwriting styles or languages.

2.7 Scope and Limitations

Scope. The study covers a translation model for the Spanish used by people during the Philippine-American War (1899–1902). It

includes OpenCV image preprocessing, text recognition on the cleaned images, and Spanish-to-English translation of the recognized text, using scanned transcript images as input.

Limitations.

- The model is trained and tested only on period Spanish from the Philippine-American War and is not meant for modern Spanish or other historical periods.
- Documents in Tagalog, other Philippine languages or English are not covered, even when they appear in the same collections.
- Results depend on the quality and number of available scanned images and parallel translations, which may be small.
- Severely damaged, torn or faded pages, and unusual handwriting, may not be recognized correctly.
- The output is a machine-generated aid and should be checked by a historian or Spanish-language expert before it is used as a formal translation.

3. REVIEW OF RELATED LITERATURE

Machine translation can make historical languages more accessible by converting texts into modern languages that readers of the modern world can understand. This benefits researchers, students, historians, and the public by giving them access to linguistic and cultural information that would otherwise need a deeper understanding of this field.

[1] The study conducted by Sobhy et al. (2023) demonstrates how artificial intelligence can be applied to historical writing. They developed a system that performs these tasks: identify and recognize hieroglyphic symbols from images and translate them into English. They used image processing and computer vision techniques to detect and recognize hieroglyphics in scanned images. However, their study ran into difficulties caused by limited data, complexity of symbols, and uneven representation of glyphs. [3] The study conducted by Miyagawa (2023) focuses on translating between Japanese and Ainu, an endangered indigenous language of northern Japan. The study used the Marian neural machine translation framework and works with existing digital texts and translations. The Japanese-Ainu system achieved a SacreBLEU score of 32.90, which shows the potential of machine translation with low resources. The two studies reveal that machine translation depends on the availability and quality of training data and the characteristics of the language.

These findings are relevant to historical Spanish because older Spanish transcripts can contain archaic spellings, specialized vocabulary, and expressions whose meanings differ from their modern counterparts. As a result, a translation model trained on modern Spanish may not reliably translate historical Spanish.

Neural machine translation uses neural networks to learn the relationship between source language and target language from examples of translated text. Pretrained models already contain some linguistic representations by learning from large datasets, which can be adapted to language pairs through additional training. [4] The study of Puranik et al. (2021) involves an investigation on fine-tuning a pretrained neural machine translation model for English-Marathi translation. They explored using parallel training data to adapt the current model to the translation task. The results achieved according to the BLEU showed promising results, indicating potential for pretrained models to provide a useful foundation for translation for languages with limited resources. The study suggests that pretrained models reduce the amount of training required to develop a translation system. However, fine-tuning depends on the quality and quantity of the additional data. Small or inconsistent data may not reveal accurate results.

[2] The study conducted by Khan et al. (2024) investigates the application of artificial intelligence and machine learning on historical transcripts and examines how OCR technologies can assist in converting images of historical documents into machine readable text. OCR technologies assisted researchers efficiently by accelerating the process since manual transcription requires a substantial amount of time and expertise in the field. They used four AI/ML approaches to process selected historical datasets: Convolutional Neural Networks (CNNs), Sequence-to-Sequence Contrastive Learning (SeqCLR), Vision Transformers, and Transformer-based OCR (TrOCR). However, accuracy depends on the quality of the documents. Therefore, it is important to evaluate the transcription quality before using historical texts for analysis.

Before text can be recognized, the image itself has to be prepared, because a degraded scan limits everything that follows. Boudraa et al. (2017) [7] addressed this for historical documents with a multi-stage binarization framework. It applies CLAHE to improve contrast in poor images, combines several existing thresholding methods to separate the text from the background, and then removes small noise and corrects character shapes. They tested the approach on three popular datasets and reported accurate, efficient results compared with well-known methods. Their work shows that contrast enhancement and thresholding are practical first steps for old manuscripts, which supports the preprocessing stage of this study.

Once an image is cleaned, the text must be recognized. Li et al. (2023) [8] proposed TrOCR, an end-to-end model that uses a pre-trained image Transformer to read the image and a pre-trained text Transformer to generate the text. The model can be pre-trained on large synthetic datasets and fine-tuned on labeled data, and it outperformed earlier state-of-the-art models on printed and handwritten text. Because Khan et al. (2024) [2] also examined TrOCR among their approaches, TrOCR is a reasonable candidate for recognizing handwritten Spanish in this study, as an alternative to a standard OCR engine.

Recognition alone does not make a historical text readable. Romero et al. (2019) [5] observed that automatic transcription of historical documents had mostly produced diplomatic transcripts, which often only experts can understand because of heavy abbreviations and outdated word forms. They studied ways to train recognition models to produce modernized transcripts instead, using Spanish documents on naval commerce from the 15th to 19th centuries. Their work shows that archaic Spanish can be modernized during transcription, which is the same barrier the Philippine-American War transcripts present.

Spelling differences also affect translation. Domingo and Casacuberta (2018) [6] noted that historical documents lack a consistent spelling convention, which complicates the preservation of cultural heritage. They proposed statistical, neural and character-based machine translation approaches to adapt a document's spelling to modern standards and reported encouraging results across several historical corpora. This suggests that normalizing archaic spelling before translation can help a translation model handle period Spanish.

Taken together, these studies show that each stage has working solutions on its own: image enhancement [7], text recognition [2], [5], [8], spelling normalization [6], and translation with limited data [1], [3]. However, none of them combines these stages into one workflow for the handwritten Spanish of the Philippine-American War. This gap is what the present study

addresses.

4. METHODOLOGY

4.1 Research Design

This study employs a quantitative experimental and comparative research design with qualitative evaluation. It aims to develop and evaluate a three-stage pipeline for translating historical Spanish transcription images into modern English. The pipeline consists of image preprocessing using OpenCV, text recognition via optical character recognition and handwritten text recognition (OCR/HTR), and neural machine translation.

4.2 Image Source

Source: Historical manuscript collection sourced through academic collaboration with the University of Nueva Caceres Museum.
Estimated Number of Images: ~44 scanned images.
Image Type: Scanned handwritten historical documents and records from the Philippine-American War period.
Resolution: High-resolution digital scans (approx. 300 DPI or higher, depending on archival preservation standards).
File Format: PNG.
Inclusion Criteria: Handwritten or printed historical Spanish transcripts/documents dating from the Philippine-American War era (1899–1902).
Exclusion Criteria: Documents written in languages other than Spanish (e.g., Tagalog or English), or pages severely damaged, torn, or faded beyond optical or manual recognition.
Licensing or Permission Considerations: Obtained through formal research coordination and permission from the museum coordinator at the University of Nueva Caceres for academic and non-commercial research use.

4.3 Image Acquisition

The dataset was acquired through institutional coordination with the University of Nueva Caceres Museum. The images consist of digital scans of archival documents preserved by museum historians. These records capture primary source materials from the historical period, featuring period-specific handwriting, ink degradation, and paper textures typical of 19th-century administrative or military correspondence. The files were securely transferred and stored in a standardized PNG format to maintain visual fidelity for subsequent OpenCV preprocessing and text recognition steps.

5. OpenCV TECHNIQUES

This section describes the image input and the OpenCV techniques the study plans to apply before text recognition. All processing will be written in Python with OpenCV in a Jupyter Notebook. The settings given here are starting values and will be tuned during the experiments described in Section 6.

5.1 Image Input

The input will be the scanned Spanish transcript images described in Section 4.2, an estimated 44 PNG files. Each image will be loaded with cv2.imread and checked for a valid path and size. OpenCV loads color images as three-channel BGR arrays, and the pipeline will work on a single grayscale channel. The original files will not be changed. Each processing step will be saved as its own image, so the effect of every technique can be compared with the raw (unprocessed) scan. Scan resolution may differ between images, so it will be recorded for each file, and pixel-based settings will be adjusted if the sizes vary widely.

5.2 Preprocessing Techniques

Step	OpenCV	Purpose
1. Grayscale Conversion	cv2.cvtColor	Keep one channel, since color is not needed for recognition
2. Noise Removal	cv2.fastNlMeansDenoising	Remove speckles and paper grain while keeping pen strokes
3. Lighting Normalization	cv2.dilate, cv2.medianBlur,	Estimate the paper background and divide it out to flatten

	cv2.divide	shadows, yellowing and stains
4. Contrast Enhancement	cv2.createCLAHE	Darken faded ink without washing out the page
5. Deskewing	cv2.getRotationMatrix2D, cv2.warpAffine	Straighten tilted pages
6. Thresholding	cv2.threshold, cv2.adaptiveThreshold	Separate text from background; global, Otsu and adaptive Gaussian will be compared
7. Cleanup	cv2.morphologyEx	Remove tiny isolated specks

Two versions of the cleaned image will be kept: a grayscale version (after deskewing) and a binarized version (after thresholding). Recognition engines may not prefer the same input, so both will be tested (Section 6).

5.3 Output of this Stage

This stage will produce (1) one image per technique, which form the preliminary processed images, (2) the grayscale and binarized versions of each page, and (3) the line boxes and one cropped image per text line. Items (1) and (2) will be shown as before-and-after figures in the paper, and item (3) will be passed to text recognition.

6. EXPERIMENTAL COMPARISON

This section plans the comparisons the study will run. Each comparison changes one factor while the rest of the pipeline stays fixed, and the results will be reported in tables and figures so that they are easy to read. No results are reported yet.

6.1 Preprocessing and Thresholding (Comparisons A and B)

The raw scan will be compared with the output of each technique in Table 1, added one at a time in the planned order, to see what each step contributes. The three thresholding methods will then be compared on the same deskewed image. Because the purpose of preprocessing is readable text, each version will also be passed to the same OCR engine, and the change in character error rate (CER) will be recorded together with the visual comparison.

6.2 Text Recognition (Comparison C)

Tesseract with Spanish language data will be used as the OCR engine and TrOCR [8] as the HTR engine, following the options examined by Khan et al. [2]. Both will read the same line crops from the binarized and the grayscale versions, and Tesseract will also be run on whole pages and on the raw scan. Accuracy will be measured with CER and word error rate (WER). A strict score will compare the text exactly, and a relaxed score will ignore case, accents and punctuation, to show how much of the error is only accents and capital letters. Processing time will also be recorded. The released TrOCR model was trained on English handwriting, so lower accuracy on Spanish accents is expected, and fine-tuning on the transcribed lines will be considered if time allows.

6.3 Translation (Comparison D)

A pretrained Marian translation model for Spanish to English will serve as the baseline, since Marian has been used for low-resource translation [3]. The following factors will be compared:

1. Spelling modernization. Translation of the text as written (raw) compared with translation after a rule-based table of period spellings has been applied, as suggested by Domingo and Casacuberta [6].

2. Translation unit. Lines joined into passages compared with each line translated alone, since a manuscript line break is not always a sentence break.

3. Input text. A perfect transcription compared with the OCR and HTR outputs, to measure how much recognition errors reduce translation quality.

4. Model. The pretrained model compared with a model fine-tuned on period Spanish and English pairs, if enough parallel text can be prepared, because the benefit of fine-tuning depends on the quality and amount of the added data [4].

Translations will be scored against the reference translation with BLEU and chrF, computed with SacreBLEU as in [3]. Historians or Spanish-language experts will also review a sample of the output for adequacy and fluency, and the rating scale will be decided with them.

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